

THEORY AND METHODS SIMULTANEOUS LOG-LINEAR MODELLING OF COMPETING RISKS

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Summary: A common approach to the analysis of survival data with multiple causes of failure is to model, separately, each cause-specific hazard rate and estimate the corresponding parameters by single-cause methods. The literature also contains an approach where two or more cause-specific hazard rates are modelled simultaneously and the parameters are estimated from the same model. In the present paper we provide analytical justification to demonstrate that these two approaches lead to the same estimates of hazard rates. More importantly, we bring to the surface the fact that simultaneous modelling is based on the assumption of independence among the causes of failure in which case it is impossible to distinguish between crude- and net-hazard rates.

1. Introduction

In the analysis of survival data with multiple causes of failure interest has centred on the rate of failure associated with each unique cause when other causes are controlled for. The usual goal is estimation of cause-specific rates and covariate effects on such rates given the presence or ‘removal’ of certain other causes.

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A common procedure to the analysis of such data is to estimate the parameters of a given cause-specific hazard rate, separately, using single-failure methods. In so doing, individuals failing due to causes other than that of interest are treated as censored at the point of completion. A sample of applications based on this procedure includes Edin (1989), Han and Hausman (1990), and Narendranathan and Stewart (1991; 1993).

In few other applications an approach has been used where two or more causes of failure are modelled simultaneously and the cause-specific parameters are estimated from the same model. Larson (1984), Gomez de Leon and Potter (1989), and Liefbroer (1991) are representative examples following such procedure.

According to Allison (1984), the type-specific hazard functions are defined in such a way that it is unnecessary to make any assumptions about dependence or independence among different causes of failure. We know from results of Lagakos (1979), however, that standard methods for estimating the hazard are based on the assumption that censoring times are independent of failure times. This, in turn, implies that the different causes being modelled act independently. As to the methods based on simultaneous modelling, Larson (1984) holds that it is not necessary to make the assumption of independence among the cause of failure though, as we shall see later, such assumption is implicit in his procedure. An immediate consequence of the independence assumption is that the net hazards are equal to the corresponding crude hazards (Elandt-Johnson and Johnson, 1980).

A number of studies have explored the feasibility of the assumption of independence as well as the consequences of wrongly making such assumption. Gail (1975), for instance, shows how different models for the

survival function lead to different estimates of partial-crude probabilities. Slud and Byar (1988) and Di Serio (1997) reveal that the problem of “unexpected protectivity” in competing-risks models, a problem that occurs when a covariate shows a protective impact not expected from a medical perspective, is due to the lack of independence between the competing failures. Using an example from family demography Hill, Axinn, and Thornton (1993) conclude that the independence assumption leads to substantially biased estimates of net marriage and net cohabitation survival functions. Klein and Moeschberger (1987) have also shown that one could be appreciably misled by falsely assuming independent component lifetimes.

Due to such problem some investigators have attempted to model multiple causes of failure in a dependent-risks framework. See for instance Klein and Moeschberger (1984), Carling and Jacobsson (1995), Yashin, Manton and Stallard (1986), Nour (1981), and Robertson and Uppuluri (1984). Other more recent works that make use of the simultaneous modelling approach or address issues related to competing risks include Boracchi, Biganzoli, and Marubini (2001), McKeague, Gilbert, and Kanki (2001), Fine and Gray (1999), Wohlfahrt, Andersen, and Melbye (1999), Ghilagaber (1998), Heisey and Nordheim (1995), Mulder and Wagner (1993), Pierce, and Preston (1993), Ansell and Phillips (1989), Schuluchter and Jackson (1989), and Gray (1988). However, it is impossible to distinguish empirically models in which different kinds of events are dependent from a model in which they are independent.

In the present paper, we further explore the approach based on simultaneous modelling of multiple causes of failure and provide analytical justification to demonstrate that it leads to estimates of hazard rates that are identical to analogous estimates obtained by using the standard approach based on separate modelling. We show, further, that existing methods of simultaneous modelling,

typically Larson (1984) and the works that followed it, do not exclusively model the crude hazard rate as they claim to without explicitly admitting that the crude and net hazards are identical under the independence assumption.

In Section 2 we define various types of hazard rates while in Section 3 we describe a log-linear model for the hazard rate, derive its likelihood functions under separate and simultaneous modelling of two or more causes of failure, and demonstrate the equivalence of the resulting estimates. The last section summarizes the contents of the paper.

2. Various hazard rates

The hazard rate (function) usually denoted by $\lambda(t)$ represents the instantaneous rate at which an event occurs at a specific point t of a (non-negative) time variable T given that it has not occurred before such time:

$$\lambda(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t < T \leq t + \Delta t | T \geq t)}{\Delta t}. \quad (1)$$

In situations where individuals are exposed to R ($R \geq 2$) causes of failure $C_1, \dots, C_R$ indexed by r ($r = 1, \dots, R$), let C represent the cause of failure. A cause-specific hazard function, $\lambda_r(t)$, may then be written as

$$\lambda_r(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t < T \leq t + \Delta t; C = r | T \geq t)}{\Delta t}, \quad r = 1, \dots, R \quad (2)$$

so that $\lambda_r(t)$ is the instantaneous rate of occurrence of the event due to cause r at a point t of the time variable T and in the presence of $R - 1$ other causes. When an individual experiences an event, the time-to-event and the cause are recorded. Some questions that arise in situations of this type are: (1) what is the distribution of the time-to-event from different causes acting

simultaneously, (2) what is the hazard that an individual fails from specific cause, say C_r , and (3) how might the failure pattern change if a certain cause, C_v , were 'removed'. Prentice et al. (1978) shed light on some of these issues.

It is supposed that each individual is endowed with a set of 'times due to experience an event' for each cause. Let $T_1, \dots, T_R$ denote these hypothetical times. The *multivariate hazard component* with respect to cause C_r , at time point $\mathbf{t} = (t_1, \dots, t_R)$ is defined as (Erlandt-Johnson and Johnson, 1980)

$$\lambda_r(\mathbf{t}) = \lambda_r(t_1, \dots, t_R) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} P \left\{ \bigcap_{s \neq r}^R (T_s > t_s) \cap (t_r < T_r \leq t_r + \Delta t) \mid \bigcap_{s=1}^R (T_s \geq t_s) \right\}. \quad (3)$$

Thus, $\lambda(\cdot) = \{\lambda_1(\cdot), \dots, \lambda_R(\cdot)\}$ is a *multivariate hazard vector*. Since each event is due to a single cause, we cannot observe $(T_1, \dots, T_R)$ jointly. Instead, we observe the time at event $T = \min(T_1, \dots, T_R)$. Since the events are mutually exclusive, the *overall (from any cause) hazard function* is

$$\lambda_T(t) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} P \left\{ \bigcap_{s=1}^R (t < T_s \leq t + \Delta t) \mid \bigcap_{s=1}^R (T_s \geq t) \right\} \quad (4)$$

which, of course, is a function of a single variable t . We can also identify cause-specific hazard rates. The instantaneous rate of failure from cause C_r , when all causes are operating simultaneously in the population is called the *crude hazard rate* and is defined as

$$\lambda_r(t) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} P \left\{ (t < T_r \leq t + \Delta t) \mid \bigcap_{s \neq r}^R (T_s \geq t) \mid \bigcap_{s=1}^R (T_s \geq t) \right\}. \quad (5)$$

It is equal to the multivariate hazard rate component evaluated at the point $\mathbf{t} = (t, t, \dots, t)$. In other words, $\lambda_r(t) = \lambda_r(t, t, \dots, t)$.

Similarly, the instantaneous rate of failure from cause C_r , when one cause (C_v , say) is 'removed' while all other causes are operating simultaneously in the population, is called the *partial-crude hazard rate* and is defined as

$$\lambda_{r|v}(t) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} P \left\{ (t < T_r \leq t + \Delta t) \bigcap_{s \neq r, v}^R (T_s \geq t) \mid \bigcap_{s \neq v}^R (T_s \geq t) \right\}. \quad (6)$$

On the other hand, the *net hazard rate* defined as

$$h_r(t) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} P \{ (t < T_r \leq t + \Delta t) \mid T_r \geq t \} = \lambda_r(0, 0, \dots, 0, t, 0, \dots, 0) \quad (7)$$

represents the instantaneous rate associated with the hypothetical time T_r , of failing due to cause C_r when C_r is the only operating cause.

Put in words, the crude hazard rate, $\lambda_r(t)$, of a cause of failure refers to a rate computed from the original model with all causes operating, while the net hazard rate, $h_r(t)$, refers to a rate computed from a model in which all other causes are 'removed'. Removal of other causes of failure consists in replacing the corresponding hazard by zero while the hazard of interest remains unaffected, irrespective of whether it would be possible to carry out the partial removal operation in real life. In general $\lambda_r(t) \neq h_r(t) \neq \lambda_{r|v}(t)$. Equality between any pair holds only when the times to failure from different causes are independent.

At this stage it may be worth noting that in the human population the net and partial-crude hazards cannot be estimated directly, but only through their relations with the crude hazard. The study of these relations is part of the problem of competing risks, or multiple decrement and has been addressed a long time ago. Formulae expressing the relations between net

and crude probabilities have been developed by assuming either a constant intensity of risk of death or a uniform distribution of deaths. Chiang (1968, Chapter 11) reviews these formulae assuming a constant relative intensity and discusses approaches of estimating net and partial-crude probabilities using these relations.

Before winding up this section it may be worth remarking that the various hazard functions described above may also be defined via the corresponding joint distribution or survival functions. The joint survival function of $T_1, \dots, T_R$ is defined as

$$S_{1,\dots,R}(t_1, \dots, t_R) = P \left\{ \bigcap_{r=1}^R (T_r > t_r) \right\} \quad (8)$$

while the marginal survival functions for each cause are defined as

$$S_r(t_r) = S_{1,\dots,R}(0, \dots, t_r, \dots, 0). \quad (9)$$

The *multivariate hazard component* with respect to cause C_r , at time point $\mathbf{t} = (t_1, \dots, t_R)$ given in (3) may then be redefined as

$$\lambda_r(\mathbf{t}) = -\frac{1}{S_{1,\dots,R}(\cdot)} \frac{\partial}{\partial t_r} [S_{1,\dots,R}(\cdot)] = -\frac{\partial}{\partial t_r} \log [S_{1,\dots,R}(\cdot)]. \quad (10)$$

The *overall (from any cause) survival function* is

$$S_T(t) = P(T \geq t) = P \left\{ \bigcap_{r=1}^R (T_r \geq t) \right\} = S_{1,\dots,R}(t, \dots, t). \quad (11)$$

which, again, is a function of a single variable t . Its corresponding hazard rate which is shown in (4) may then be obtained as

$$\lambda_T(t) = -\frac{1}{S_T(t)} \frac{\partial}{\partial t} [S_T(t)] = -\frac{\partial}{\partial t} \log [S_T(t)] = -\frac{\partial}{\partial t} \log S_{1,\dots,R}(t, \dots, t). \quad (12)$$

Similarly, the *crude hazard rate* defined in (5) may be redefined as

$$\lambda_r(t) = -\frac{1}{S_T(t)} \frac{\partial}{\partial t_s} [S_{1,\dots,R}(\cdot)] \Big|_{\{t_s=t\}} = -\frac{\partial}{\partial t_s} \log [S_{1,\dots,R}(\cdot)] \Big|_{\{t_s=t\}} \quad (13)$$

while the *net hazard rate* shown in (7) may be redefined as

$$h_r(t) = -\frac{\partial}{\partial t} \log [S_r(t)] = -\frac{\partial}{\partial t} \log [S_{1,\dots,R}(0, \dots, t, \dots, 0)]. \quad (14)$$

3. Modelling multiple causes of failure

3.1 A log-linear hazard model

It is possible to estimate the crude hazard rate using non-parametric methods like the Nelson-Aalen estimate of the cumulative hazard rate (Nelson, 1969; 1972; Altshuler, 1970; Aalen, 1978). In many applications, however, investigators are faced with problems of analyzing survival data grouped into a finite number of time intervals in the form of a frequency table. In such situations it is practical to assume that the baseline hazards are piece-wise constant (are constant within each time interval but may vary between intervals). This is equivalent to a piece-wise uniform distribution of events (say, deaths) and is in accordance with Chiang's (1968) assumption in formulating relationships among the various hazard types. Further, in the presence of covariates, this leads to a multiplicative (log-linear) hazards model which is a grouped-data version of the common Cox (1972) type of proportional hazards model as shown in equation (16) below (see also, Hoem, 1993; Ghilagaber,

1995). The use of loglinear models, so common in the analysis of contingency tables, in the field of survival analysis has also been demonstrated by Holford (1980), Laird and Olivier (1981), and Gilula (1986).

For these and other reasons it is assumed, in the present paper, that the overall hazard rate $\lambda_T(t)$, as well as the cause-specific hazard rates $\lambda_r(t)$, are approximated by step-functions on I disjoint time intervals indexed by i ($i = 1, \dots, I$). Further, suppose one has recorded information on a categorical covariate indexed by j ($j = 1, \dots, J$). In such discrete formulation, it is more convenient to use λ_i instead of $\lambda(t)$, for the hazard rate at time interval i .

A log-linear model for such a hazard rate arises when we express the log-hazard as a linear function of the time interval and the categorical covariate:

$$\ln \lambda_{ij} = \Delta + \alpha_i + \beta_j, \quad i = 1, \dots, I; j = 1, \dots, J \quad (15)$$

or, equivalently

$$\lambda_{ij} = \exp(\Delta + \alpha_i + \beta_j) = \exp(\Delta + \alpha_i) \exp(\beta_j) = \lambda_{0i} \exp(\beta_j), \quad i=1, \dots, I; j=1, \dots, J \quad (16)$$

where λ_{ij} is the rate at which an individual with the j^{th} level of the covariate experiences an event in interval i .

The notation and procedure used above is analogous to the analysis of variance on the log-hazards. There is a grand mean Δ , and main effects α_i and β_j (no interaction is assumed). To avoid overparameterization (ensure a unique solution), it is necessary to make certain restrictions on the parameters. Procedures in standard programs like SPSS, SAS and SPlus achieve this by setting one of the parameters, often that corresponding to the highest level, equal to zero ($\alpha_I = \beta_J = 0$).

The assumption that the hazard rate $[\lambda_i = \lambda(t_i)]$ is approximated by step-functions on I disjoint time intervals t_i indexed by i , implies that the time-to-event variable T follows what is known as a piece-wise exponential distribution with parameter λ_i . Its density function is then given by

$$f(t_i) = \lambda_i \exp(-\lambda_i t_i). \quad (17)$$

Let w_{ijm} be an indicator variable of whether the m^{th} sample member having the j^{th} level of the covariate experiences the event of interest ($w_{ijm} = 1$) or is censored ($w_{ijm} = 0$) in the i^{th} time interval. The contribution, to the likelihood, of the subsample in the i^{th} time duration and having the j^{th} level of the covariate can then be obtained as

$$\Lambda_{ij} = \prod_m (\lambda_{ij})^{w_{ijm}} \exp(-\lambda_{ij} t_{ijm}) = (\lambda_{ij})^{D_{ij}} \exp(-\lambda_{ij} T_{ij}), \quad (18)$$

where D_{ij} is the number of j -valued individuals who experience the event (say deaths) in interval i , and T_{ij} is the aggregate exposure time of all j -valued individuals in the interval i .

The likelihood for the entire sample consisting of $(I \times J)$ cells will then be given by

$$\Lambda = \prod_i \prod_j \Lambda_{ij} = \prod_i \prod_j (\lambda_{ij})^{D_{ij}} \exp(-\lambda_{ij} T_{ij}) \quad (19)$$

Final estimates of the parameters in (15) may be obtained by first replacing λ_{ij} in (19) by the right-hand side of (16) and then maximizing (19). The procedure can easily be extended to the case of more than two covariates. For details, see Laird and Olivier (1981), Larson (1984) or Ghilagaber (1995).

3.2 Separate log-linear modelling of cause-specific hazard rates

Assume that there are two causes of failure. Further, assume, as before, that we have records of one categorical covariate. Let the corresponding hazard rates at interval i be denoted by $\lambda_{ij}^{(1)}$ and $\lambda_{ij}^{(2)}$, where

$$\lambda_{ij}^{(r)} = \exp \left[\Delta^{(r)} + \alpha_i^{(r)} + \beta_j^{(r)} \right], \quad r = 1, 2; i = 1, \dots, I; j = 1, \dots, J. \quad (20)$$

Let the recorded exposures in group (i, j) be T_{ij} , and let there be $D_{ij}^{(1)}$ and $D_{ij}^{(2)}$ occurrences of the event due to causes 1 and 2, respectively. Then the likelihoods are [by analogy to (19)]

$$\Lambda_1 = \prod_i \prod_j (\lambda_{ij}^{(1)})^{D_{ij}^{(1)}} \exp(-\lambda_{ij}^{(1)} T_{ij}), \quad i = 1, \dots, I; j = 1, \dots, J, \quad (21)$$

and

$$\Lambda_2 = \prod_i \prod_j (\lambda_{ij}^{(2)})^{D_{ij}^{(2)}} \exp(-\lambda_{ij}^{(2)} T_{ij}), \quad i = 1, \dots, I; j = 1, \dots, J. \quad (22)$$

The type-specific parameters on the right-hand side of (20) can therefore be estimated by substituting (20) in (21) and (22) and maximizing, *separately*, the corresponding likelihoods as usual.

3.3 Simultaneous log-linear modelling of cause-specific hazard rates

If we combine (21) and (22), the total likelihood, *assuming independence between the two causes of failure*, becomes

$$\Lambda = \Lambda_1 \Lambda_2 = \prod_{r=1}^2 \prod_i \prod_j (\lambda_{ij}^{(r)})^{D_{ij}^{(r)}} \exp(-\lambda_{ij}^{(r)} T_{ij}), \quad i = 1, \dots, I; j = 1, \dots, J. \quad (23)$$

If there was only one cause of failure and two categorical covariates indexed by r ($r = 1, 2$) and j , then the corresponding likelihood function would have looked as follows:

$$\prod_{r=1}^2 \prod_i \prod_j (\lambda_{rij})^{D_{rij}} \exp(-\lambda_{rij} T_{rij}), \quad i = 1, \dots, I; j = 1, \dots, J \quad (24)$$

where T_{rij} represent the total exposure time of all (r, j) -valued individuals experiencing an event in interval i . The only essential difference between (23) and (24) is that (24) contains two exposures T_{1ij} and T_{2ij} while (23) contains only one set of exposures T_{ij} though it appears twice, namely both in the product $\lambda_{ij}^{(1)} T_{ij}$ and $\lambda_{ij}^{(2)} T_{ij}$. If we denote T_{1ij} and T_{2ij} by T_{ij} , say, and further write λ_{rij} for $\lambda_{ij}^{(r)}$ and D_{rij} for $D_{ij}^{(r)}$, then (23) may be rewritten as

$$\prod_{r=1}^2 \prod_i \prod_j (\lambda_{rij})^{D_{rij}} \exp(-\lambda_{rij} T_{rij}), \quad i = 1, \dots, I; j = 1, \dots, J. \quad (25)$$

This demonstrates that (23) and (24) are identical except for notational differences. In such a case, it is possible to analyze both types of failure *simultaneously* by 'fooling' computer programs such as SAS, SPSS and SPlus into believing that the cause of failure [indexed by r in (24)] is another factor operating with the 'real' factors. Such a procedure has been used earlier by Larson (1984), De Leon and Potter (1989), and Liefbroer (1991). However, none of these earlier works includes analytical justification as outlined above or reference to any earlier such work.

Moreover, in all previous applications including Larson (1984), the occurrences D_{rij} and exposures T_{rij} for an event of interest have been calculated by assuming that individuals experiencing any of the other events

are censored at the time of completion. The likelihood function arising from such a procedure is just the product of the marginal likelihoods corresponding to each cause. The tacit assumption of this procedure is that the different causes of failure are independent, in which case the *net* and *crude* hazards are equal. Larson (1984), however, claims that the assumption of independence is not crucial and that the procedure yields exclusively estimates of parameters corresponding to *crude* hazards.

3.4 Estimates of relative and baseline hazards

From the results of simultaneous modelling relative hazards (hazards relative to that of the baseline level) corresponding to levels of covariates may be estimated as follows: For the covariate indexed by j let j_0 be its baseline level (the default in most programs is $j_0 = J$, which is the highest level). Then, the relative hazard (due to cause r) at the j^{th} level of the covariate ($j = 1, 2, \dots, J - 1$) relative to that at level $j_0 = J$ is denoted by $b_j^{(r)}$ and is given by

$$b_j^{(r)} = \frac{\lambda_{ij}^{(r)}}{\lambda_{ij_0}^{(r)}} = \exp \{ (\beta_j - \beta_{j_0}) + [(\varepsilon\beta)_{rj} - (\varepsilon\beta)_{rj_0}] \} \quad (26)$$

for $r = 1, 2, \dots, R$; and $i = 1, 2, \dots, I$. Note that $r = 1$ refers to the events due to cause 1, $r = 2$ refers to the events due to cause 2, etc., and ε stands for the effect of the pseudo-variable *Cause-of-failure* on log-hazards, while $(\varepsilon\beta)$ stands for an interaction between this pseudo-variable and the real covariate indexed by j . Analogous formulae may be used in the presence of more than two covariates. These relative hazards may or may not be constant across the various causes under investigation.

The baseline hazards (there is one baseline hazard for each cause of failure) are computed as follows. Let $a_i^{(r)}$ denote the baseline hazard of the r^{th} cause at the i^{th} age group. Then (for $r = 1, \dots, R$, and $i = 1, \dots, I$)

$$a_i^{(r)} = \exp \left\{ \Delta^{(r)} + \varepsilon_r + \alpha_i^{(r)} + \beta_{j_0}^{(r)} + (\varepsilon\alpha)_i^{(r)} + (\varepsilon\beta)_{j_0}^{(r)} \right\}. \quad (27)$$

From the estimates of baseline and relative hazards one may compute estimates of cause-specific hazard rates at any combination of levels of the covariates:

$$\lambda_{ij}^{(r)} = a_i^{(r)} b_j^{(r)}, \quad r = 1, \dots, R; i = 1, \dots, I; j = 1, \dots, J. \quad (28)$$

Further, from the baseline hazards one may compute, among other things,

$$\frac{a_{ij_0}^{(1)}}{a_{ij_0}^{(2)}} = \exp \left\{ [\varepsilon_1 - \varepsilon_2] + [(\varepsilon\alpha)_i^{(1)} - (\varepsilon\alpha)_i^{(2)}] + [(\varepsilon\beta)_{j_0}^{(1)} - (\varepsilon\beta)_{j_0}^{(2)}] \right\}. \quad (29)$$

These ratios will be constant if the relative frequency of events (experienced by the baseline group) due to the various causes is the same across the different age groups (values of i). On the other hand, these ratios vary across the age groups if the baseline levels do, for instance, experience more events due to cause 1 than due to cause 2 at a certain age group while they experience events due to cause 2 more often than events due to cause 1 at other age groups.

4. Discussion

Simultaneous modelling of multiple causes of failure has been used earlier. Previous applications of simultaneous modelling have, however, been based on the assumption of independence among the causes of failure, either without

explicitly specifying such assumption or holding that such an assumption is not necessary. In the actual analysis, however, times ending in events other than that of interest have been treated as censored. Since censoring is assumed to be noninformative (failure times are independent of censoring times) it is implicit that the different causes of failure are independent, in which case the net and crude hazards are equal. Thus, the current literature on simultaneous modelling of multiple causes of failure is based on a likelihood function that is simply the product of likelihood functions corresponding to the marginal likelihoods.

The present paper has attempted to extend the existing literature by providing analytical justification to demonstrate that existing simultaneous models lead to estimates of relative and absolute hazards that are identical to those obtained from the standard approach based on separate modelling. By doing so, we have also demonstrated the fact that current approaches of simultaneous modelling are based on the assumption of independence among the different causes of failure where it is impossible to distinguish between the net and crude hazard rates. As a byproduct we have also demonstrated that maximization of the total of two or more likelihoods leads to the same results (in terms of relative and baseline hazards) as maximizing the individual hazards. This, of course, is in accordance with previous explanations though the implicit assumption of independence was not clearly stated earlier.

On the basis of our analytical results we recommend that proper care be taken to identify the type of hazard being modelled.

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